

# XGBoost For Classification

## Prerequisite & Disclaimer

- 1) XGBoost Regression
- 2) Gradient Boosting classification
- 3) Log Coods)

## Problem statement

| CGPA | Placed |
|------|--------|
| 5.70 | 0      |
| 6.25 | 1      |
| 7.10 | 0      |
| 8.15 | 1      |
| 9.60 | 1      |

Stage 1 - Base Estimator

$$\log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

$$= \log\left(\frac{3/5}{2/5}\right) = \log\left(\frac{3}{2}\right)$$

$$= 0.405$$

$$\text{Prob} = \frac{e^{\log \text{odds}}}{1 + e^{\log \text{odds}}} = \frac{e^{0.405}}{1 + e^{0.405}} = 0.60$$

| Cgpa | Placed | Pred 1 | Prob | res 1 | (Placed - Prob) |
|------|--------|--------|------|-------|-----------------|
| 5.70 | 0      | 0.405  | 0.6  | -0.6  |                 |
| 6.25 | 1      | 0.405  | 0.6  | 0.4   |                 |
| 7.10 | 0      | 0.405  | 0.6  | -0.6  |                 |
| 8.15 | 1      | 0.405  | 0.6  | 0.4   |                 |
| 9.60 | 1      | 0.405  | 0.6  | 0.4   |                 |

Stage 2 : Decision Tree

leaf node -

-0.6, 0.4, -0.6, 0.4, 0.4  $\rightarrow$  similarity  
score

$$SS = \frac{(\sum \text{res}_i)^2}{\sum \text{Prev-Prob}(1 - \text{Prev-Prob}_i) + \lambda}$$

$$= \frac{(-0.6 + 0.4 - 0.6 + 0.4 + 0.4)^2}{5 \times [0.6(1 - 0.6)] + 0}$$
$$= 0$$

Splitting Criteria

The goal is to maximize SS

Splitting criteria = Arg of consecutive  
data points.

| cgpa               | res1 |
|--------------------|------|
| 5.97 $\Gamma$ 5.70 | -0.6 |

|      |      |      |
|------|------|------|
| 6.67 | 6.25 | 0.4  |
| 7.62 | 7.10 | -0.6 |
| 8.87 | 8.15 | 0.4  |
|      | 9.60 | 0.4  |

Splitting Criteria = 5.97

$cgpa < 5.97$

-0.6

$SS_L$

0.4, -0.6, 0.4, 0.4

$SS_R$

$$\frac{(-0.6)^2}{0.6(-1-0.6)}$$

$$= \frac{0.36}{0.24}$$

$$= 1.5$$

$$\frac{(0.4 - 0.6 + 0.4 + 0.4)^2}{4 \times 0.6 \times 0.4}$$

$$= \frac{0.36}{4 \times 0.24} = \frac{1.5}{4}$$

$$= 0.37$$

$$\text{Gain} = (SS_L + SS_R) - SS_{\text{root}}$$

$$= 1.5 + 0.37 - 0$$

$$= 1.87$$

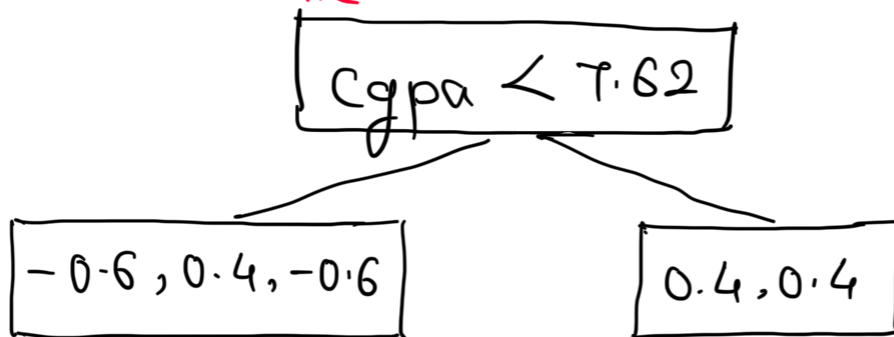
Repeating same process to obtain gain on each splitting criteria.

Gain on splitting criteria 6.67 is 0.13

Gain on splitting criteria 7.62 is 2.22.

which is max so no need to calculate for last splitting criteria

So for model 2 Decision Tree will become like



Output for a particular leaf

$$= \sum \text{residual}_i$$

$$\sum \text{Prev-prob} (1 - \text{prev-prob}) + \lambda$$

$$OP_L = \frac{-0.6 + 0.4 - 0.6}{2 \times 0.6 \times 0.4}$$

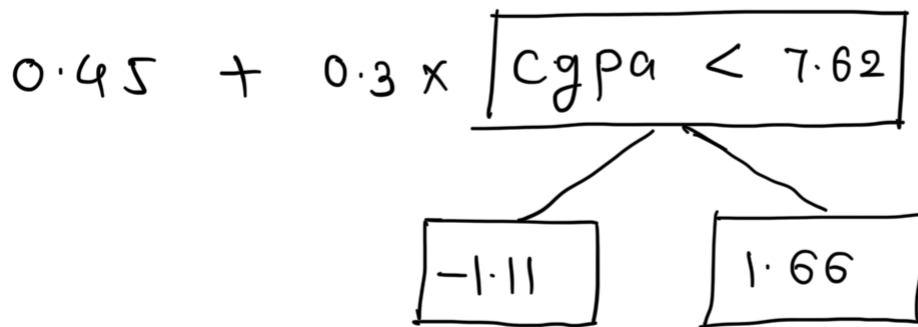
$$= \frac{-0.8}{2 \times 0.24} = -1.11$$

$$OP_R = \frac{0.4 + 0.4}{2 \times 0.24}$$

$$= 1.66$$

Stage 3: Prediction

$$m_1 + 0.3 \times m_2$$



0.3 = Learning Rate

$$\text{Pred 2} = 0.405 + 0.3 \times -1.11$$

$$= 0.072$$

| cgpa | placed | Pred1 | Prob | res1 | Pred2 | Prob  | res2  |
|------|--------|-------|------|------|-------|-------|-------|
| 5.70 | 0      | 0.405 | 0.6  | -0.6 | 0.072 | 0.518 | -0.51 |
| 6.25 | 1      | 0.405 | 0.6  | 0.4  | 0.072 | 0.518 | 0.48  |
| 7.10 | 0      | 0.405 | 0.6  | -0.6 | 0.072 | 0.518 | -0.51 |
| 8.15 | 1      | 0.405 | 0.6  | 0.4  | 0.903 | 0.712 | 0.28  |
| 9.60 | 1      | 0.405 | 0.6  | 0.4  | 0.903 | 0.712 | 0.28  |

### Stage 4 model 3

Decision<sub>3</sub>

Prediction

Prob

res3